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Make This **Handy Zener Meter** For Faster Testing

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The circuit presented here is compact and handy for measuring the voltage rating of a Zener diode instantly. To check a Zener diode, it is connected across a DC power supply with a suitable resistor in series, then the voltage drop across Zener diode is measured using a multimeter. The whole setup, including selection of voltage range in multimeter, takes more time than the measurement itself.

This circuit uses only a 9V battery as a power supply for checking voltage drop across a Zener diode that may be rated even higher than 9V. It may also be used as a continuity tester, diode junction tester, LED tester, etc. It displays basically the voltage drop across its two probes, which is useful for such testing.

Circuit and working

The circuit diagram of the Zener meter is shown in Fig. 1. It is built around a 5V voltage regulator 7805 (IC1), microcontroller ATtiny85 (IC2), an organic light-emitting diode (OLED) display, transistor 2N2222 (T1), buck-boost converter (Board1) and few other components.

The ATtiny85 microcontroller (MCU) configured as a voltmeter using its ADC channel displays the voltage across the probes on OLED display. A buck-booster module is used to step-up the 9V DC input to about 42V DC, which is required for measuring the voltage drop across the Zener diode. Due to small size of the components used here, the circuit becomes compact and handy.

The OLED display used here is 2.4cm in size with 128 × 64 pixel resolution, which requires a 5V DC supply and an I2C connector. It has four pins, out of which two are for power supply, one for serial data (SDA) and one for the clock (SCL). The text/message is displayed in 16 × 8 bit size font and voltage is displayed in 32 × 24 bit size font.

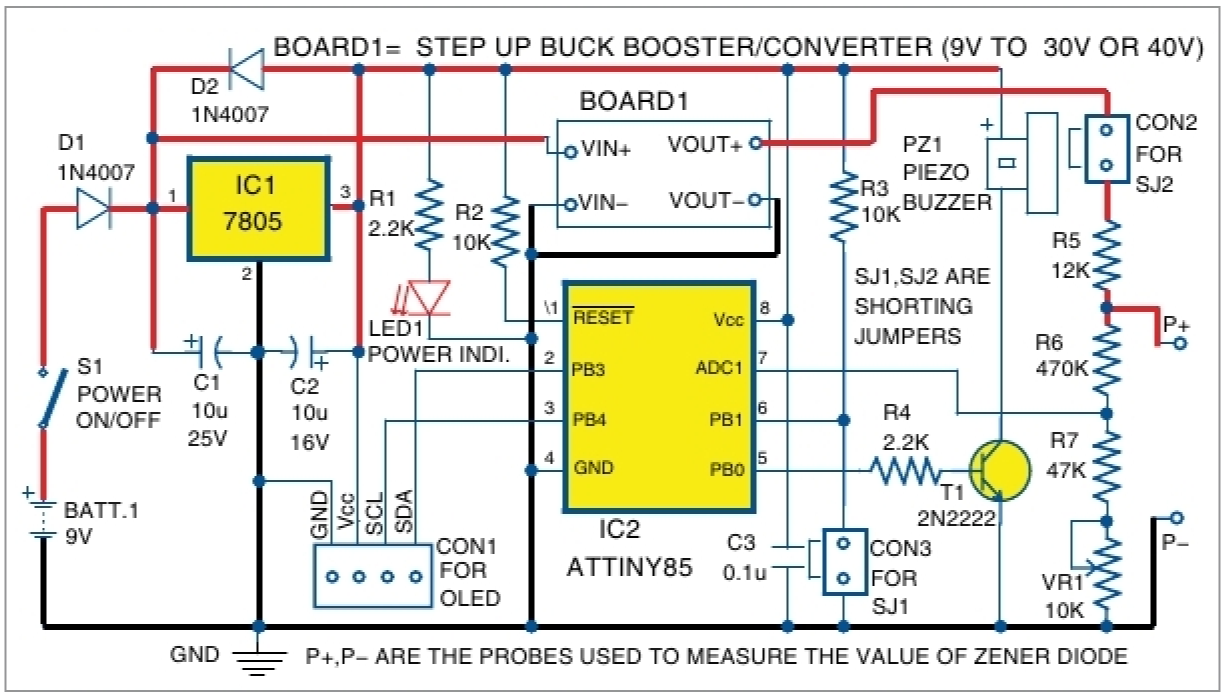

Fig. 1: Circuit diagram of handy Zener meter

A buck-boost converter (Board1) is a type of DC-DC switch-mode power supply module to change the output voltage. The buck-boost converter used here is an LM2577 based step-up module. Any step-up buck-boost module with more than 30V DC output may be used in this circuit. The circuit can work in either 0-30V range (in case jumper SJ1 is shorting) or 0-40V range (in case jumper SJ1 is open/disconnected). The range may be selected based on the maximum output of the buck-boost module.

PARTS LIST

Semiconductors:	
IC1	- 7805 voltage regulator
IC2	- ATtiny85 microcontroller
D1, D2	- 1N4007 rectifier diode
T1	- 2N2222 npn transistor
LED1	- 5mm LED
Board1	- LM2577 buck-boost converter
Resistors (all 1/4-watt, ±5% carbon):	
R1, R4	- 2.2-kilo-ohm
R2, R3	- 10-kilo-ohm
R5	- 12-kilo-ohm
R6	- 470-kilo-ohm
R7	- 47-kilo-ohm
VR1	- 10-kilo-ohm potmeter
Capacitors:	
C1	- 10µF, 25V electrolytic
C2	- 10µF, 16V electrolytic
C3	- 0.1µF ceramic disk
Miscellaneous:	
BATT.1	- 9V battery
CON1	- 4-pin connector
CON2, CON3	- 2-pin connector
S1	- On/off switch
SJ1, SJ2	- Shorting jumper
PZ1	- Piezo buzzer
	- 4-pin OLED, 2.4cm 128×64 pixel
	- 8-pin IC base
	- Jumper wires
	- Berg connector
	- 2-pin terminal connector for battery

(All these are available from kitsnspares.com)

Software

The software code (ZenerMeter.c) for ATtiny85 MCU is written in C language. It requires three libraries (oled_lib.c, font_basic_16x8.h and oled_nums_32x24.h), which are kept in the same folder as ZenerMeter.c. The hex code (ZenerMeter.hex) can be generated using any AVR C compiler such as Programmers Notepad (WinAVR). During testing, we used ISP programmer for burning the hex code ZenerMeter.hex into ATtiny85. You can use any suitable AVR MCU programmer.

The software controls the display through I2C serial communication protocol. The physical I2C pins (SDA and SCL) of ATtiny85 are not used in